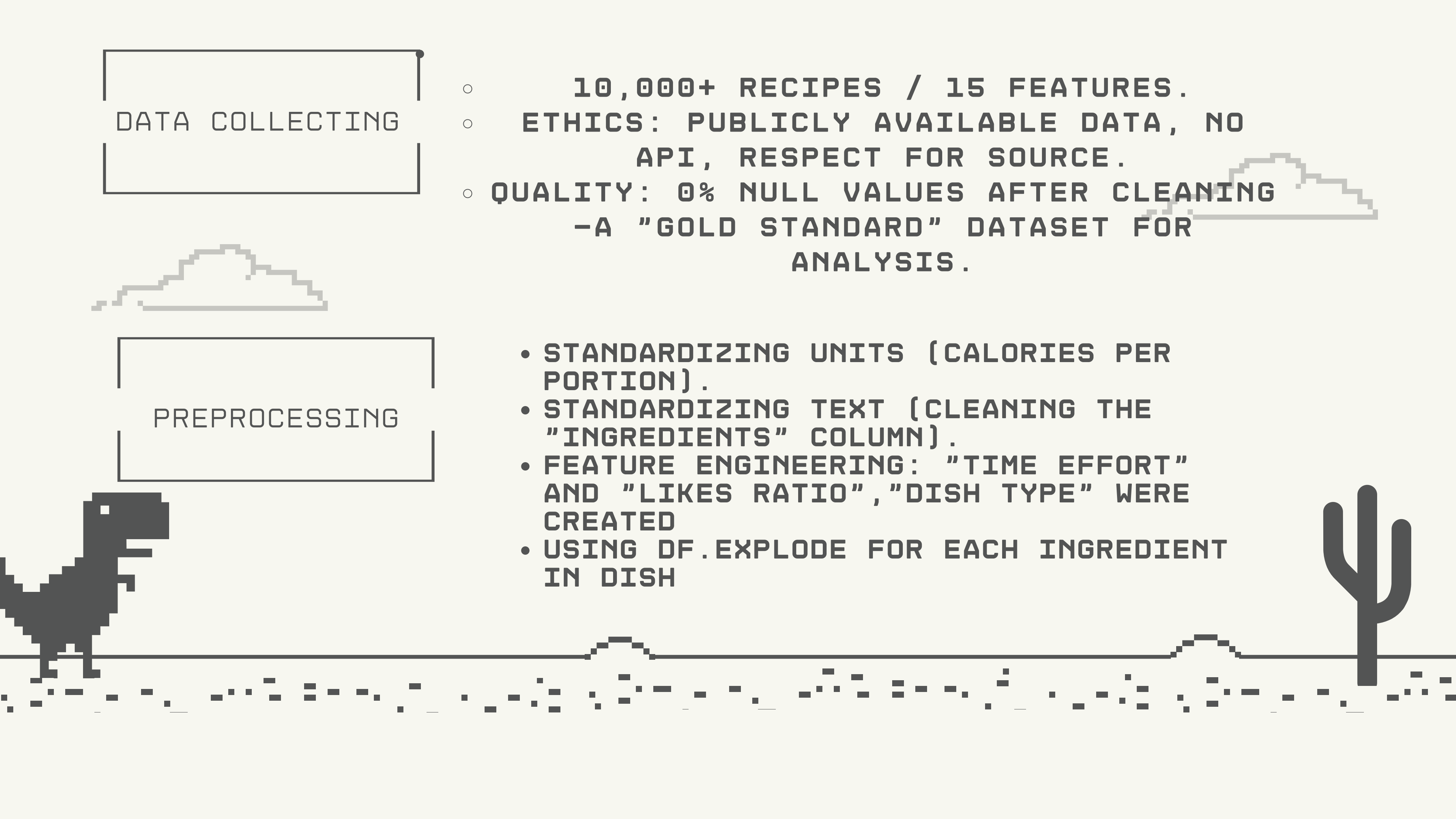


RECIPE WEBSITE ANALYSIS

DANA, AKMEIR, FAZILAT, FERUZA]

S T A R T





DATA COLLECTING

- 10,000+ RECIPES / 15 FEATURES.
- ETHICS: PUBLICLY AVAILABLE DATA, NO API, RESPECT FOR SOURCE.
- QUALITY: 0% NULL VALUES AFTER CLEANING
-A "GOLD STANDARD" DATASET FOR ANALYSIS.

PREPROCESSING

- STANDARDIZING UNITS (CALORIES PER PORTION).
- STANDARDIZING TEXT (CLEANING THE "INGREDIENTS" COLUMN).
- FEATURE ENGINEERING: "TIME EFFORT" AND "LIKES RATIO", "DISH TYPE" WERE CREATED
- USING `DF.EXPLODE` FOR EACH INGREDIENT IN DISH

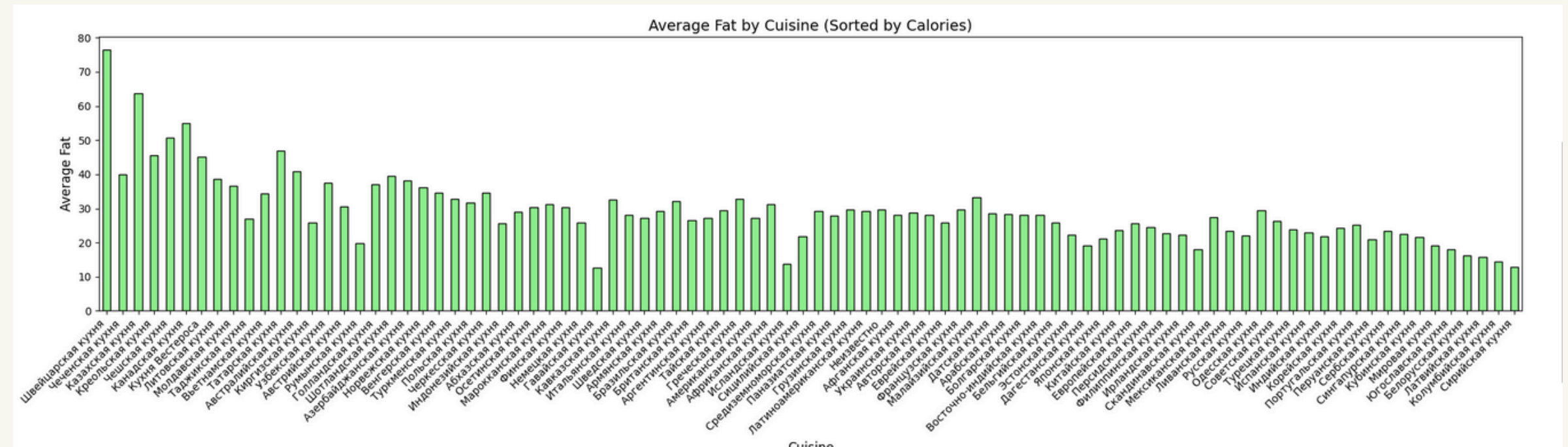
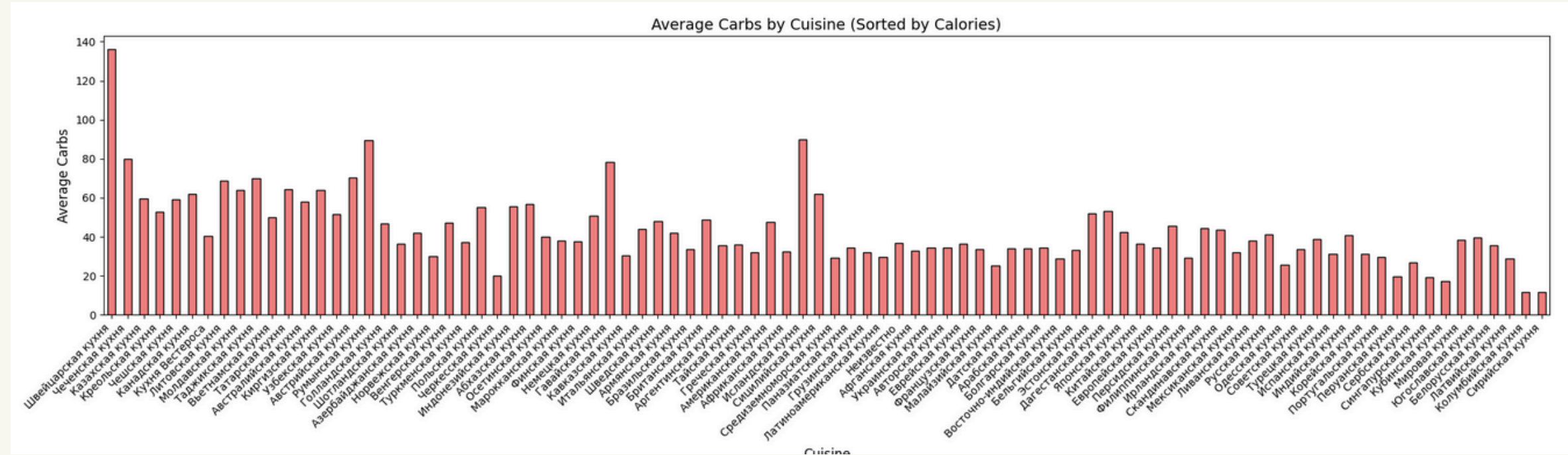
EDA

DISTRIBUTION ANALYSIS

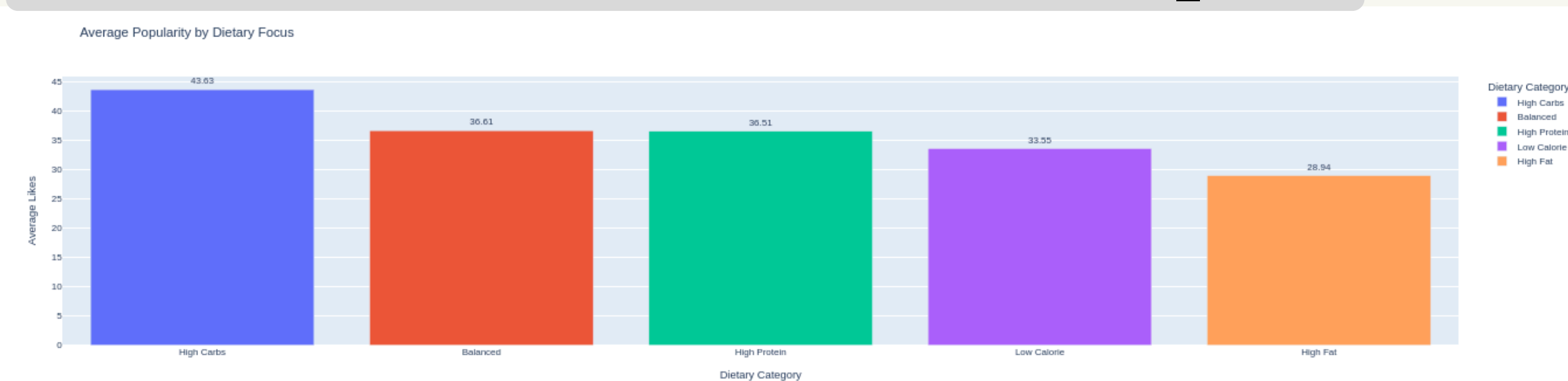
THIS LEAD US TO
THE IDEA IF THOSE
STANDING OUT
CUISINES GETS MORE
LIKES



OBSERVING THIS
DISTRIBUTIONS WE CAN
NOTICE THAT SOME
CUISINES STANDS OUT BY
THEIR EXCEPTIONALLY
HIGH AVERAGE FATS AND
CARBS.



A BAR CHART SHOWING AVERAGE LIKES PER DIETARY_FOCUS.

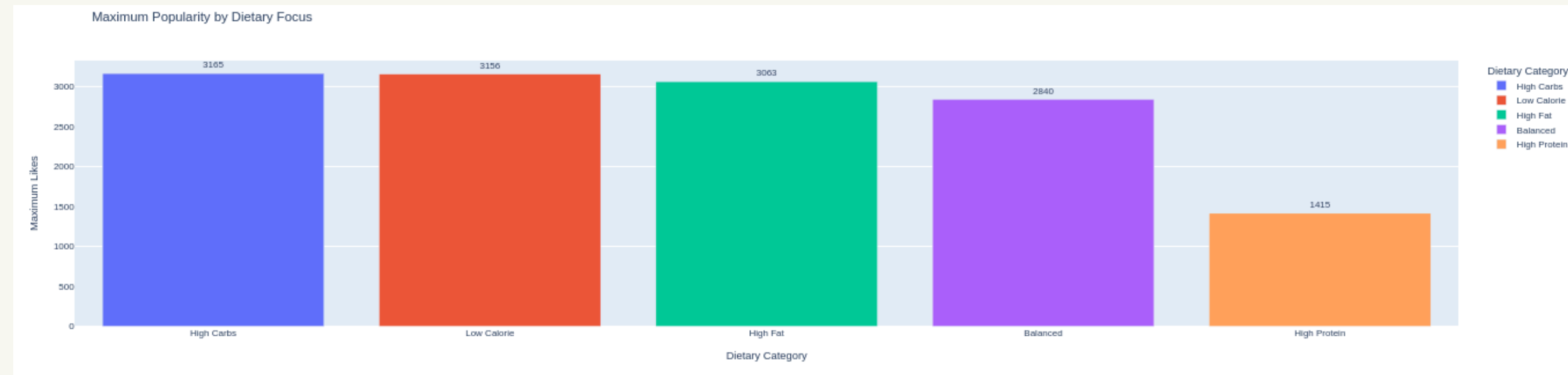


To satisfy our curiosity for the firsts we created two bar charts to distinguish between Typical User Behavior (the Average) and Viral Potential (the Maximum).

Narrative

EXPLORING THE REASONS WHICH DRIVES PEOPLE TO LIKE RECIPES AND DIFFERENT CUISINES

A BAR CHART SHOWING MAXIMUM LIKES PER DIETARY_FOCUS.



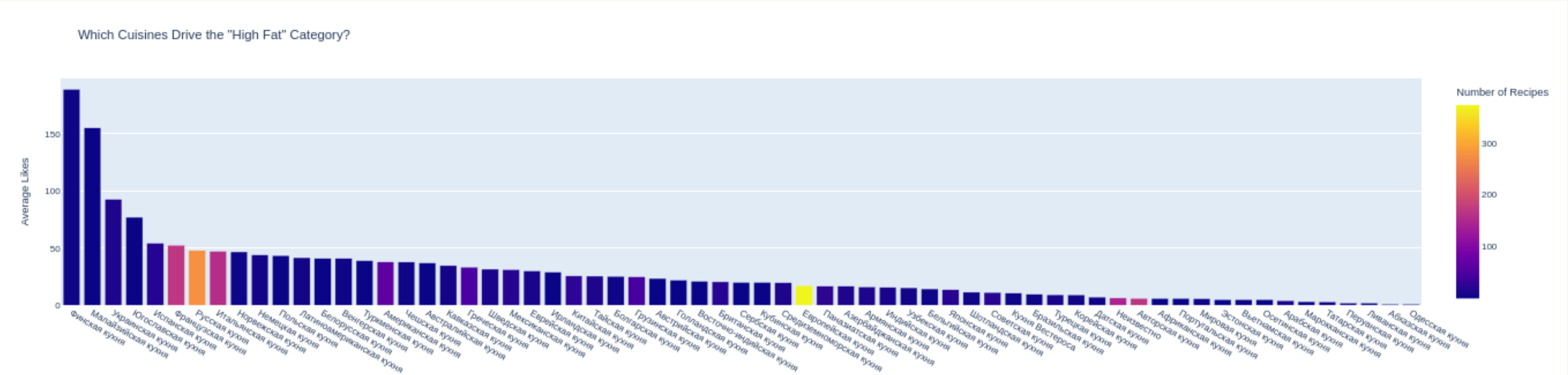
THE DEFENSE

MY INITIAL ASSUMPTION WAS THAT ALL INDULGENT (UNHEALTHY) CATEGORIES WOULD BE POPULAR. HOWEVER, THE DATA REVEALS A CLEAR PREFERENCE FOR CARBOHYDRATES OVER FATS. THIS SUGGESTS THAT USER ENGAGEMENT IS DRIVEN SPECIFICALLY BY 'COMFORT FOODS' (BREAD, PASTA, POTATOES) RATHER THAN 'HEAVY/GREASY' FOODS

HERE WE ENCOUNTERED ANOTHER INTERESTING MOMENT WHILE EXPLORING OVERALL UNHEALTHY FOOD. WE ARE SAYING THAT HIGH FATS HAS LOW RATE OF LIKES BECAUSE ITS 'HEAVY/GREASY.BUT HOW WOULD WE KNOW EXACTLY THAT ITS TRUE? LETS CHECK!



QUESTION : IF HIGH FAT FOODS HAS LOW AVERAGE BECAUSE OF IT HEAVYNESS AND OVERALL UNAPPETIZING REPRESENTATION OR SINCE "HIGH FAT" COULD ALSO MEAN HEALTHY FATS (AVOCADOS, NUTS) OR GOURMET CHEESES AND IT HAS LACK OF ATTENTION BECAUS IT'S JUST 'NICHE'



COUNTRIES THAT STANDS OUT:

- EUROPEAN CUISINE
- FRENCH CUISINE
- RUSSIAN CUISINE



WE DISCOVERED THAT POPULARITY IS DRIVEN BY ACCESSIBILITY AND CULTURAL IDENTITY. CARBS ARE 'UNIVERSAL COMFORT,' WHILE HIGH FAT IS 'NICHE GOURMET'

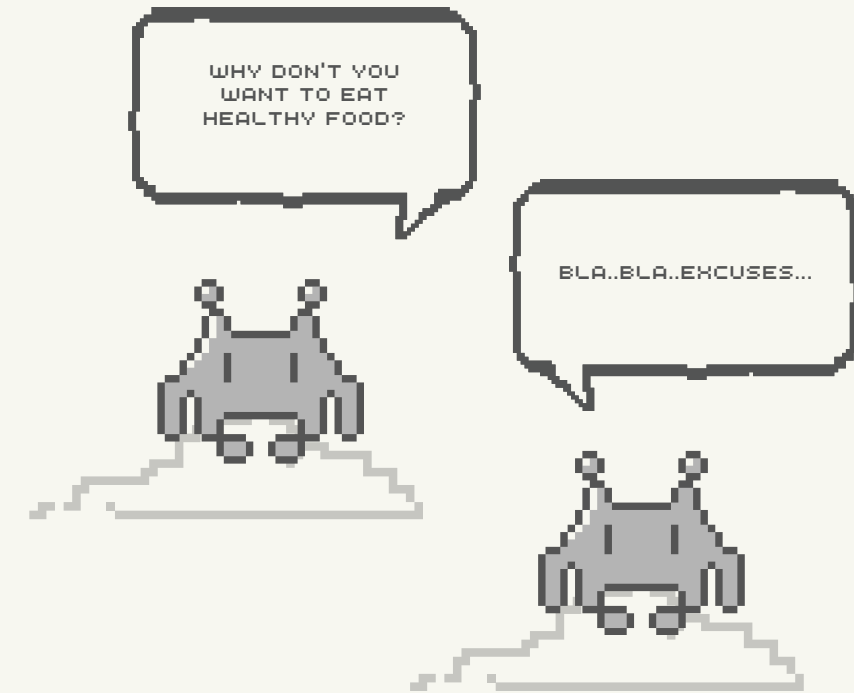
PROVE INTERESTING HYPOTHESIS

"DO HIGH-CALORIE RECIPES GET MORE LOVE THAN
HEALTHY, LOW-CALORIE ONES?"

THE "GUILTY PLEASURE": LOW-CALORIE VS. HIGH-
CALORIE.

RESULT: HIGH-CALORIE RECIPES GET 27% MORE
LIKES

INSIGHT: SOCIAL MEDIA ENGAGEMENT FAVORS
INDULGENCE OVER HEALTH.



STATISTICAL RESULTS

🖱️ P-VALUE: $P=0.003$

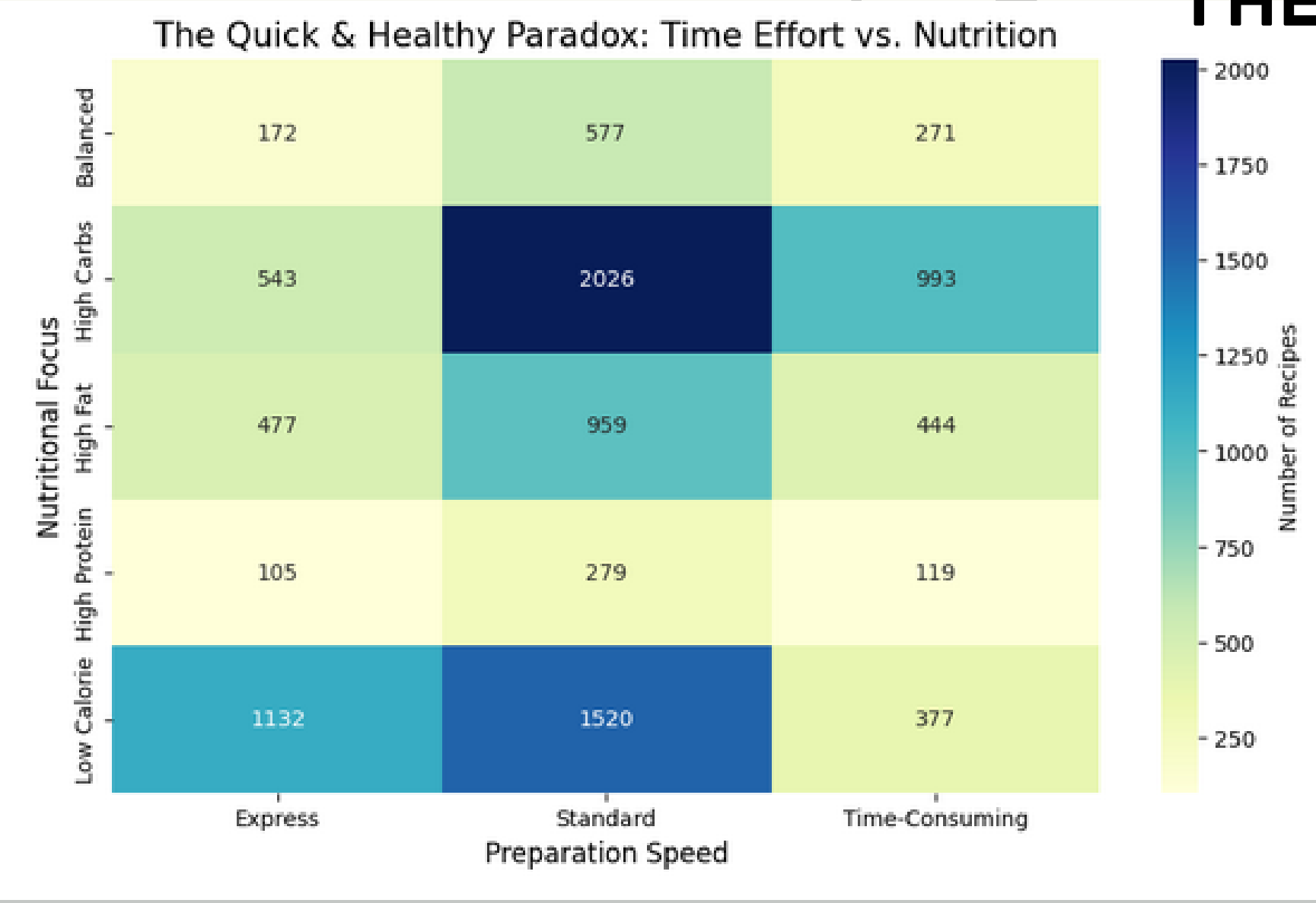
🖱️ P-VALUE < 0.05

🖱️ FAIL TO REJECT!

next

OUR DATA SHOWS THAT WHILE THE 'HEART' WANTS INDULGENCE, THE 'CLICK' FOLLOWS COMFORT AND CULTURE. TO IMPROVE PUBLIC HEALTH, WE MUST STOP FIGHTING THESE INSTINCTS AND START DESIGNING HEALTHY FOOD THAT MIMICS THE CARB-HEAVY COMFORT AND GOURMET LUXURY THAT USERS CLEARLY CRAVE ALTHOUGH NOT FORGETTING EXISTING HEALTHY FOOD ADVANTAGES

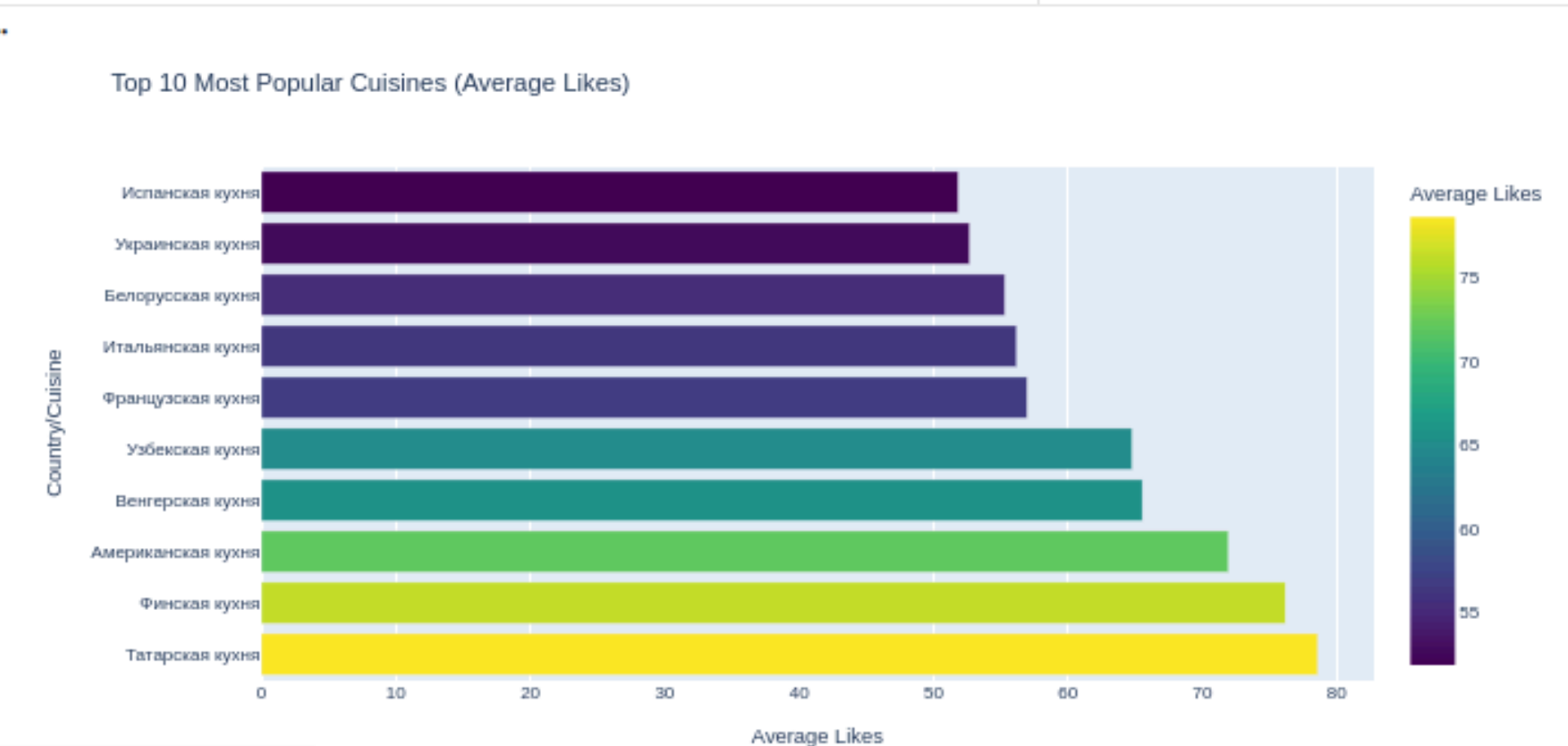
THE "QUICK & HEALTHY" PARADOX



THE NARRATIVE: DEBUNKING THE MYTH THAT HEALTHY EATING TAKES TOO MUCH TIME.

INSIGHT: LOW-CALORIE AND HIGH-PROTEIN MEALS ARE ACTUALLY FASTER TO PREPARE THAN HIGH-CARBS/FAT COUNTERPARTS.

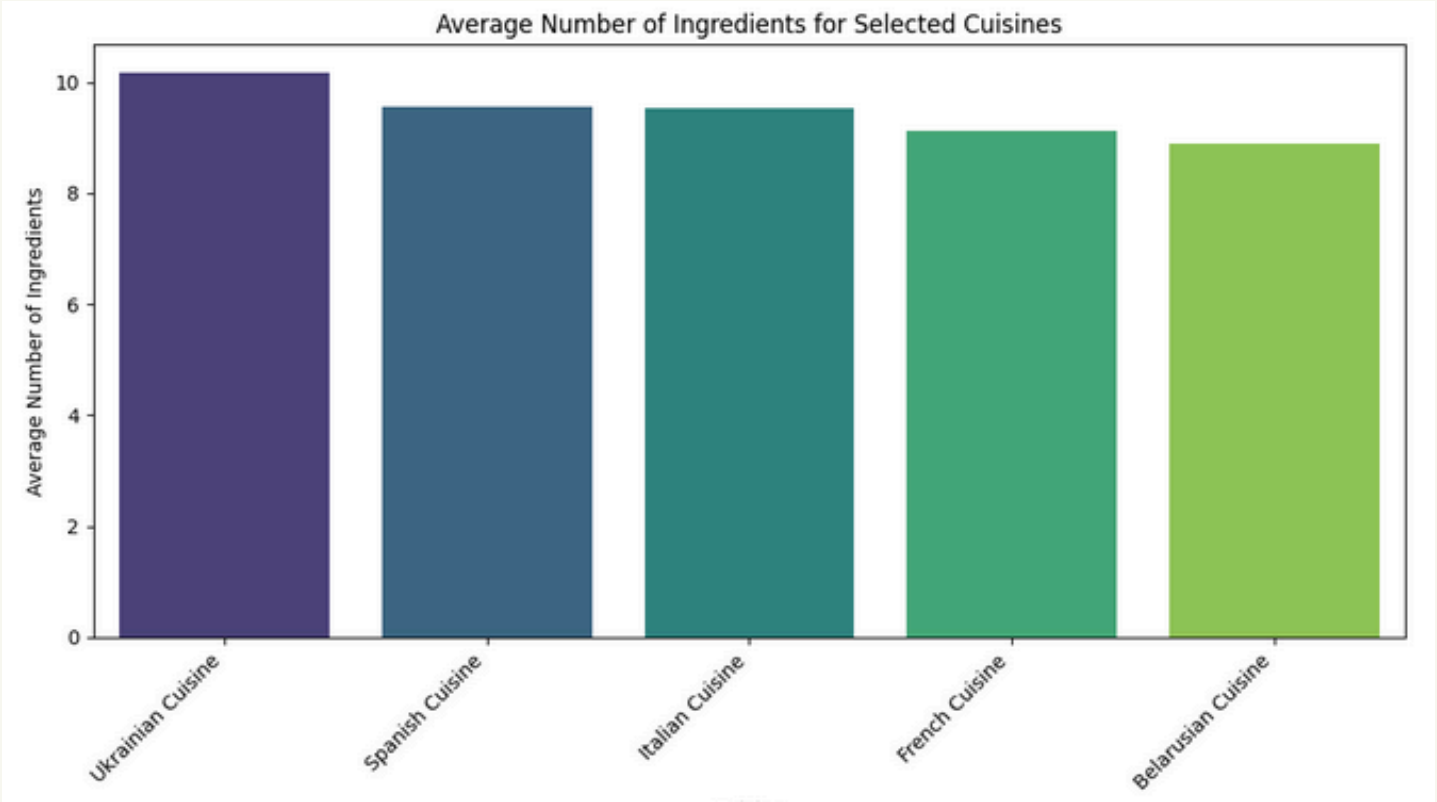




HERE PRESENTED THE MOST POPULAR CUISINES WHICH THEN WE EXPLORE BY THEIR AVERAGE NUMBER OF INGRIDIENTS AND WE WERE RIGHT THERE'S OBVIOUS RELATION SINCE IN THE TOP 5 THERE'S AVERAGE INGRIDIENTS BETWEEN 8-10 INGREDIENTS

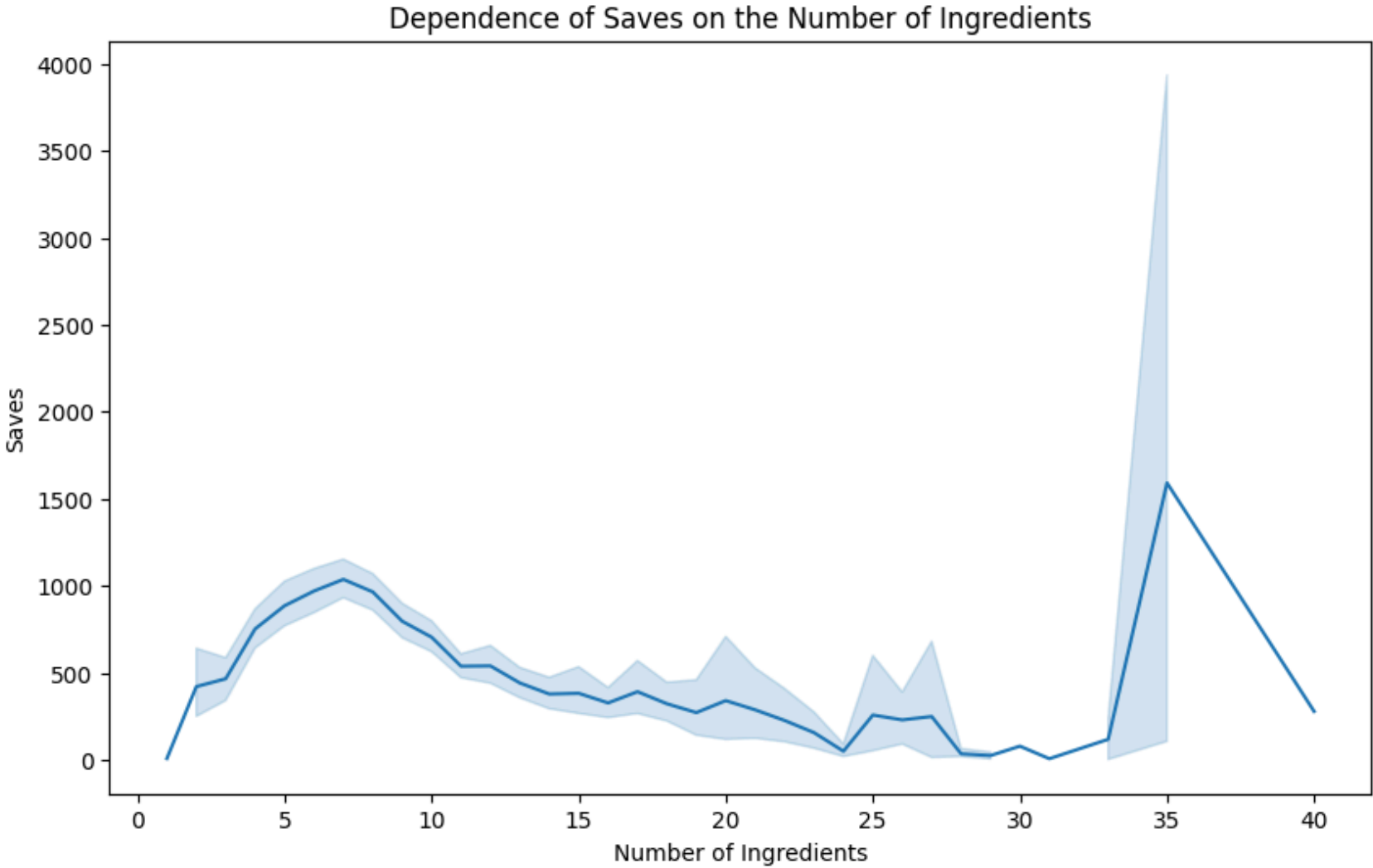
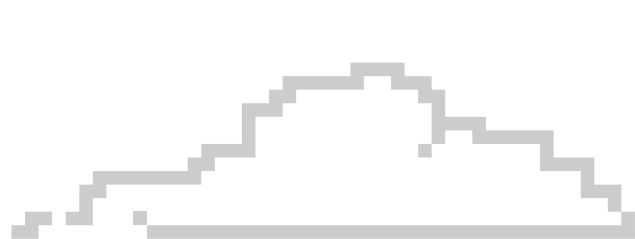
STEP1 list of top cuisines

STEP2 average number of ingredients for top cuisines



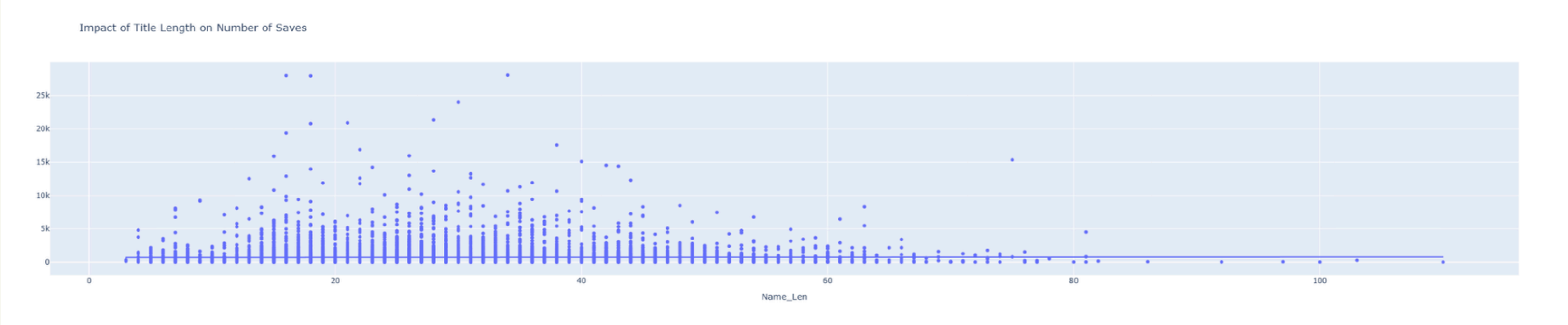
The Law of Diminishing Returns

THE NARRATIVE: COMPLEXITY SCARES PEOPLE AWAY.
INSIGHT: THE SWEET SPOT IS 8 INGREDIENTS.
THE "SPECIAL OCCASION" SPIKE: MENTION THE UNEXPECTED GROWTH AT 35+ INGREDIENTS (ELABORATE DISHES FOR WEDDINGS/EVENTS).

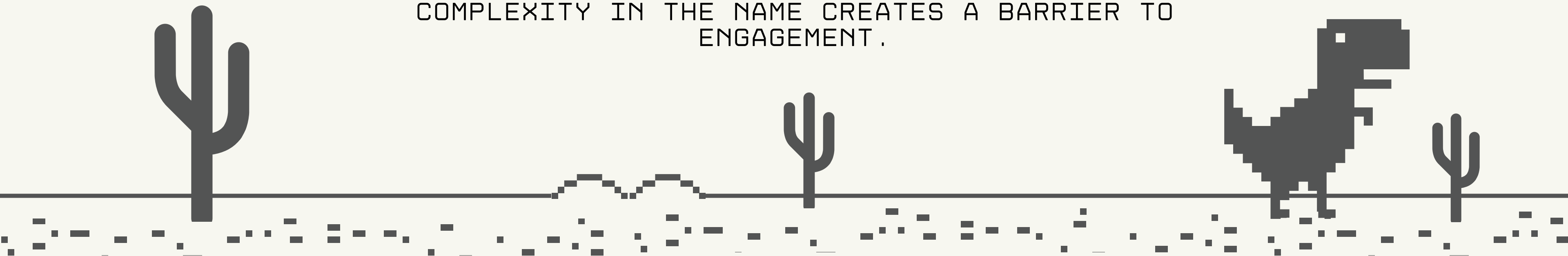


CORRELATION BETWEEN NUMBER OF INGREDIENTS AND SAVES: **-0.1327**
NUMBER OF RECIPES WITH 35+ INGREDIENTS: **1**
PERCENTAGE OF TOTAL DATASET: **0.01%**
AVERAGE SAVES FOR THE REST: **708.12**

SIMPLICITY WINS – THE TITLE & THE LIKES



THE NARRATIVE: OUR BRAIN PREFERS "BROWNIES" OVER "ARTISANAL HAND-CRAFTED COCOA SQUARES."
INSIGHT: SHORT, CONCISE TITLES DRIVE VIRALITY. COMPLEXITY IN THE NAME CREATES A BARRIER TO ENGAGEMENT.



HYPOTHESIS TESTING - CULTURE & CRAVINGS



THE "BATTLE" ITALIAN VS. RUSSIAN CUISINE

THE TEST: INDEPENDENT SAMPLES T-TEST.

THE RESULT: $P=0.014$ (THRESHOLD <0.05).

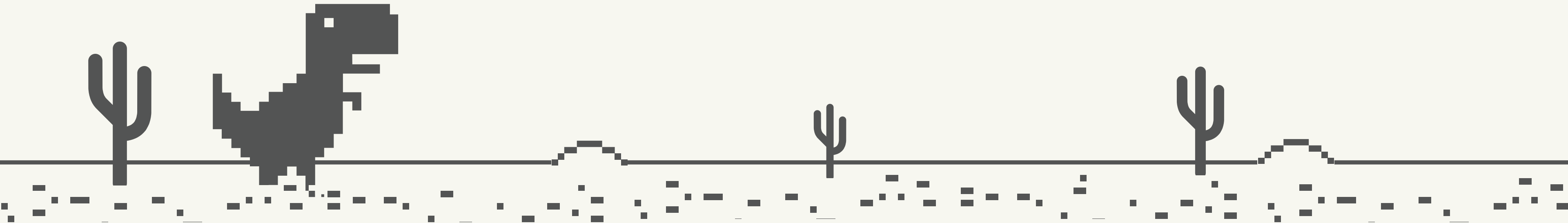
THE OUTCOME: REJECT NULL HYPOTHESIS (H_0). THE POPULARITY GAP IS REAL, NOT RANDOM.

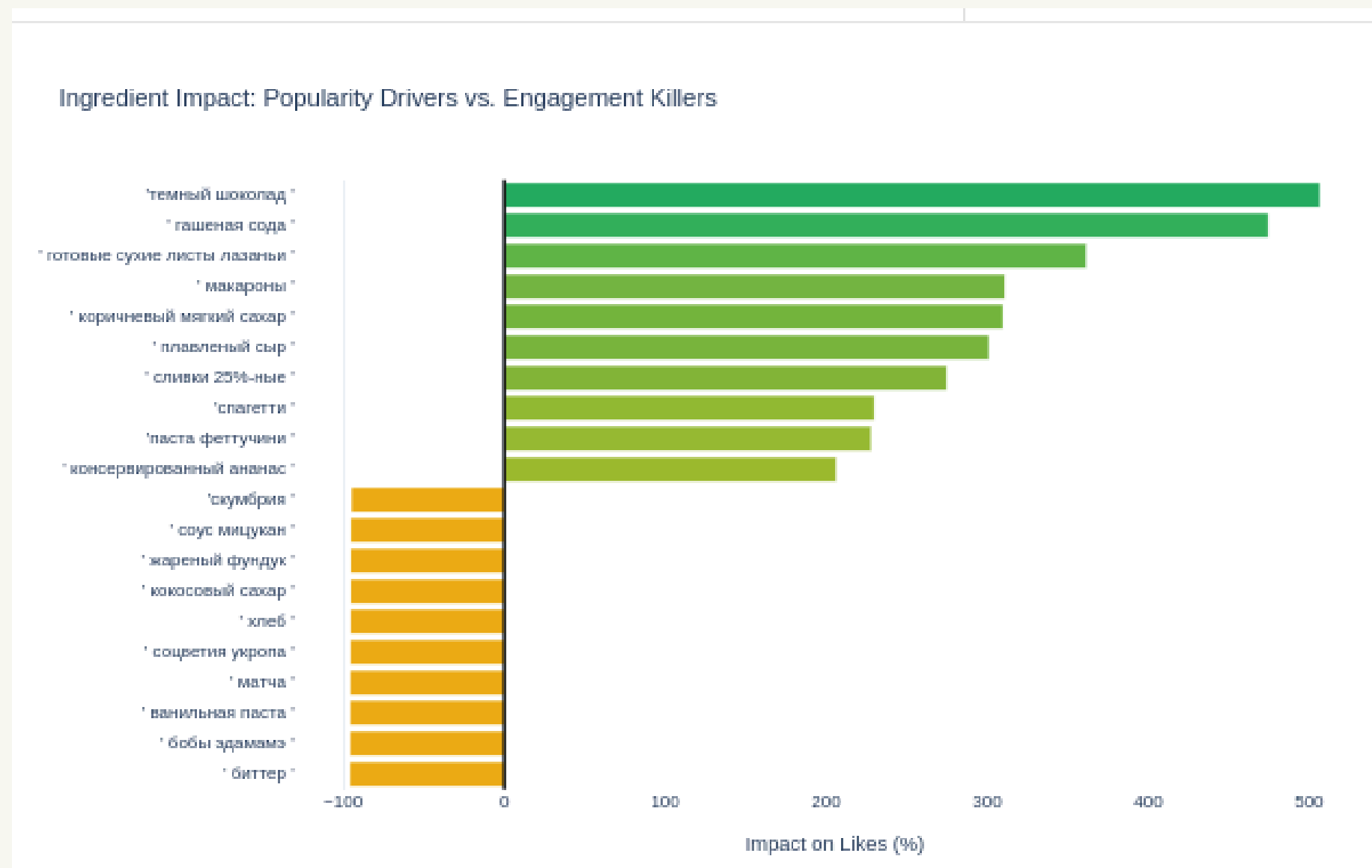
THE IMPACT: ITALIAN RECIPES GATHER 36% MORE LIKES ON AVERAGE.



ITALIAN CUISINE HAS ACHIEVED "GLOBAL RECOGNITION." IT RELIES ON A SPECIFIC SET OF HIGH-CARB, HIGH-REWARD INGREDIENTS (PASTA, CHEESE, TOMATO) THAT ARE UNIVERSALLY LOVED. RUSSIAN CUISINE, WHILE RICH (HIGH FAT), IS MORE "ACQUIRED"—IT HAS A HIGHER BARRIER TO ENTRY FOR THE AVERAGE GLOBAL USER.

FOR CONTENT CREATORS: IF YOU WANT CONSISTENT GROWTH, POST ITALIAN. IF YOU WANT NICHE AUTHORITY, POST RUSSIAN.





INGREDIENTS LIKE [ЧЕРНЫЙ ШОКОЛАД] AND [СУХИЕ ЛИСТЫ ЛАЗАНЬИ] SHOW A MASSIVE POSITIVE LIFT. CHEFS AND CREATORS SHOULD HIGHLIGHT THESE IN TITLES AND THUMBNAILS TO STOP THE USER FROM SCROLLING."

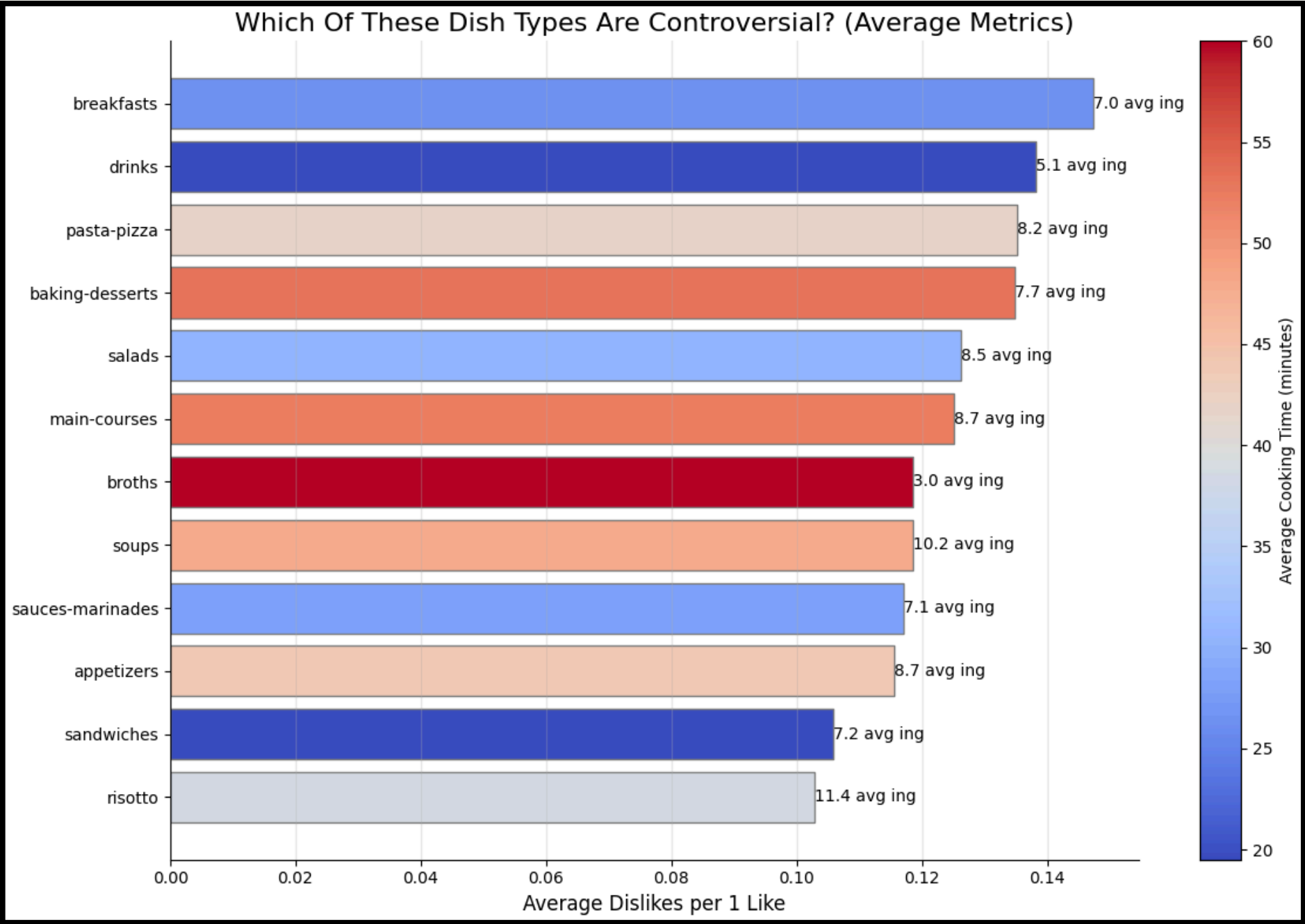


observation

IN THIS LIST OF INGREDIENTS MAKING RECIPE BECOME POPULAR, THEY'RE SEVERALS THAT ARE THE BASE FOR ITALIAN CUISINE



RISK VS. REWARD

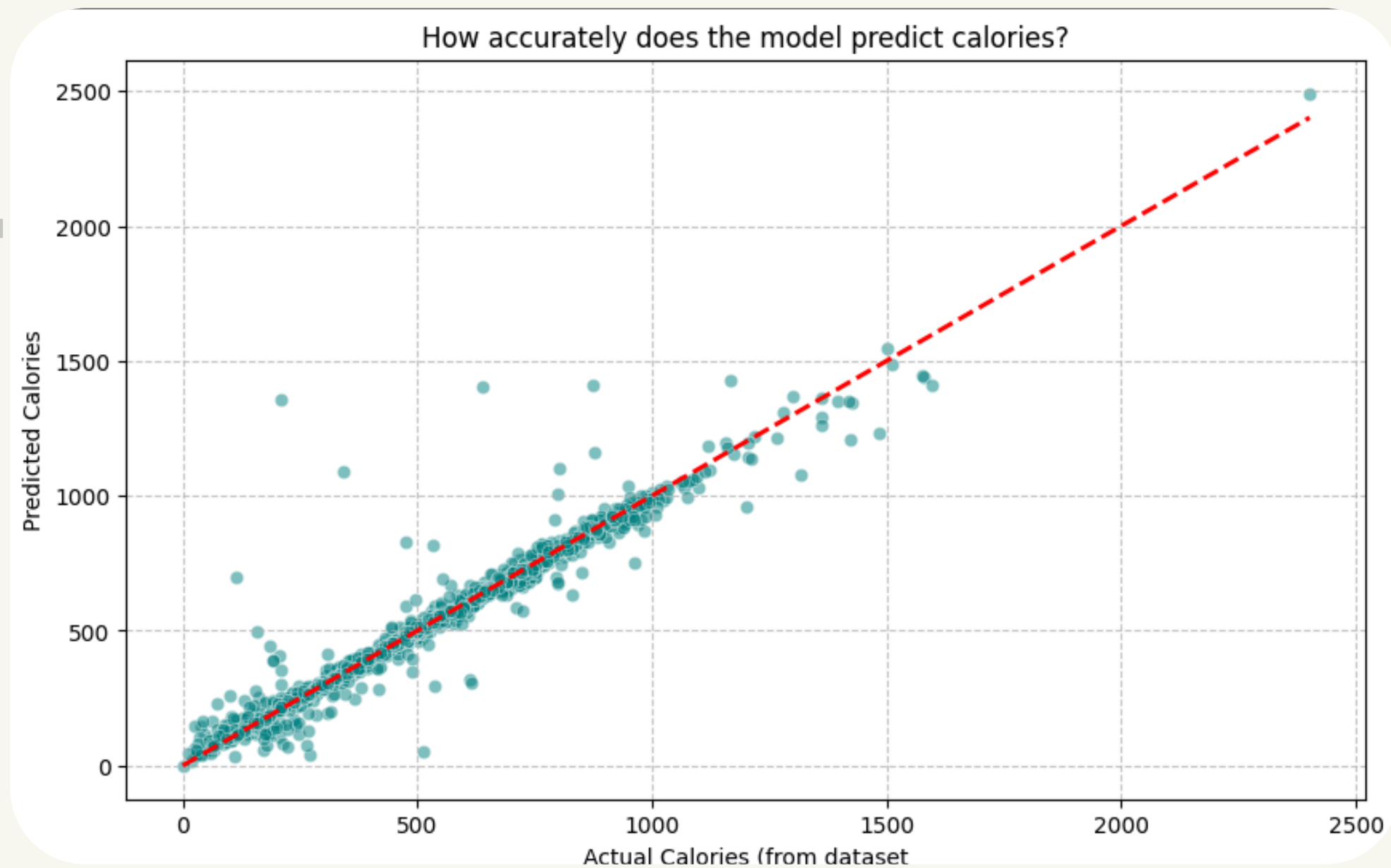


THE NARRATIVE: IF PREVIOUS GRAPHS WAS ABOUT WHAT PEOPLE LIKES THIS ONE IS ABOUT WHAT PEOPLE TOLERATE.

POLARIZING DISHES. BREAKFASTS AND DRINKS ARE THE MOST "CONTROVERSIAL." WHY? LIKELY DUE TO HIGH EXPECTATIONS FOR TASTE/QUALITY IN "SIMPLE" CATEGORIES.

CALORIE PREDICTION (ML MODEL 1)

THE NARRATIVE: CAN WE GUESS THE CALORIES IF WE ONLY KNOW THE MACROS?



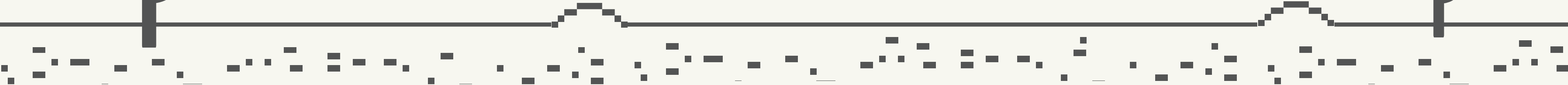
- **ACCURACY:** $R^2=0.96$.
- **THE DRIVER:** FAT IS THE #1 PREDICTOR (69% IMPORTANCE).
- **VALUE:** THIS MODEL CAN ACT AS AN INSTANT CALORIE CALCULATOR FOR UNTAGGED RECIPES.

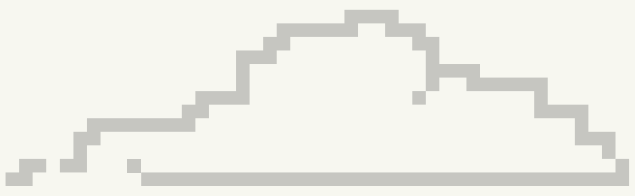
SEMANTIC SEARCH - TEACHING AI TO "TASTE"



THE NARRATIVE: MOVING BEYOND KEYWORDS TO UNDERSTANDING "INTENT."
THE CHALLENGE: YOUR MODEL FOUND "LIQUEUR" WHEN LOOKING FOR "HEALTHY BREAKFAST."
INSIGHT: WHILE THE TECHNICAL IMPLEMENTATION (SENTENCETRANSFORMER) IS PERFECT, THE "DATA GAP" WAS REVEALED. THE DATASET IS CURRENTLY WEIGHTED TOWARD DRINKS/BAR MENUS.
LESSON: AI IS ONLY AS GOOD AS THE DIVERSITY OF THE TRAINING DATA.

Query	Top Score	Quality of Results
ПП завтрак с яйцами	0.82	Excellent. Found poached eggs, Eggs Benedict, and "syrupey" breakfasts.
быстрый ужин с курицей	0.81	Perfect. Found quick chicken fillets, chicken pilaf, and chicken soup.
завтрак для детей	0.72	Strong. Identified overnight oats and family-style sandwiches.





RECOMMENDATIONS

FOR CONTENT CREATORS & CHEFS

- THE "8-INGREDIENT RULE"
 - REBRAND HEALTHY OPTIONS
 - FOCUS ON TITLES
- 



FOR PLATFORM DEVELOPERS (EDA.RAMBLER.RU)



AUTOMATED CALORIE TAGGING
DYNAMIC SEARCH FILTERING
PERSONALIZATION
DYNAMIC SEARCH FILTERING
PERSONALIZATION